HSMMamba: Hierarchical Superpixel MoE With Mamba for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification aims to fully exploit spatial and spectral information for fine-grained land-cover recognition in complex scenes. However, existing methods still face two major challenges. On the one hand, traditional convolutional networks are ineffective at modeling long-range spatial–spectral dependencies, whereas self-attention-based models usually suffer from high computational complexity. On the other hand, although hierarchical superpixels provide effective multiscale structural priors, most existing methods rely on fixed cross-level mapping relationships and shared feature transformation strategies, making them inadequate for handling the pronounced heterogeneity among features at different scales. To address these issues, this article proposes a hierarchical superpixel mixture-of-experts Mamba network for HSI classification, termed HSMMamba. The proposed framework unifies hierarchical structural priors, adaptive mixture-of-experts enhancement, and global–local joint state-space modeling within a single architecture. In particular, a learnable hierarchical superpixel mapping (LHSM) module is first developed to adaptively optimize cross-level downsampling and upsampling relationships by introducing learnable residuals on top of fixed hierarchical mapping priors, thereby enhancing the consistency of multilevel feature propagation and reconstruction. Then, a hierarchical superpixel mixture-of-experts module is introduced to perform differentiated modeling of pixel-level and multiscale superpixel-level features, and to adaptively fuse them through a gating mechanism, thus improving the representation capability of heterogeneous multilevel features. Finally, a forward–backward–differential Mamba module is constructed to preserve the strength of long-range dependence modeling while enhancing the characterization of local boundaries, fine-grained variations, and class transition regions. Extensive experiments demonstrate that HSMMamba achieves overall accuracies (OAs) of 98.98%, 99.30%, and 95.65% on the three public HSI datasets of Pavia University (PU), Salinas (SAs), and HongHu, respectively, verifying its effectiveness and superiority in complex scenarios.**

***Index Terms*—Deep learning, hyperspectral image (HSI) classification, Mamba, small sample, superpixel.**

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I. Introduction

HYPERSPECTRAL images (HSIs), which acquire land-cover information over continuous and densely sampled spectral bands, possess rich spectral resolution and have therefore demonstrated significant application value in land-cover mapping [1], [2], precision agriculture [3], [4], environmental monitoring [5], and urban analysis [6]. Compared with conventional RGB images, HSIs not only provide more detailed spectral response information but also preserve complex spatial structural characteristics, thereby offering a more informative basis for fine-grained land-cover recognition [7], [8]. Nevertheless, HSI classification remains a challenging task. On the one hand, different classes often exhibit high spectral similarity [9], while samples within the same class may show considerable intraclass variability due to illumination changes, shadow interference, and background mixing [10], [11]. On the other hand, complex scenes commonly involve blurred boundaries, densely distributed small targets, and pronounced transition regions, which further increase the difficulty of HSI classification [12], [13].

In recent years, deep learning has significantly advanced HSI classification by enabling end-to-end spatial–spectral feature learning. Existing methods can generally be grouped into CNN-based, GCN-based, and Transformer-based approaches [14], [15], [16], [17], [18], [19]. CNN-based methods are effective in extracting local spatial–spectral patterns, GCN-based methods model non-Euclidean relations among pixels or regions, and Transformer-based methods capture global contextual dependencies through self-attention. Representative studies include SpectralFormer [20], CLOLN [21], the attention multihop graph, and multiscale convolutional fusion

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network [22], MorphFormer [23], SSFTT [24], and GSCViT [25]. In addition, domain adaptation and few-shot learning methods have also been explored to improve cross-domain robustness and data-efficient representation learning [26], [27].

Although these methods have achieved promising performance, they still suffer from several limitations. CNNs are constrained by local receptive fields and are insufficient for modeling long-range spatial–spectral dependencies [28], [29], [30]. Transformers provide strong global contextual modeling, but the self-attention mechanism usually introduces high computational and memory costs for high-dimensional hyperspectral data [31], [32], [33], [34]. GCNs are sensitive to graph construction quality and mainly rely on local neighborhood aggregation, which may limit their ability to capture complex spatial–spectral correlations and long-range dependencies [35], [36], [37], [38]. These limitations motivate the development of a more efficient framework for global dependence modeling in HSI classification [39], [40].

Recently, Mamba, a selective state-space model, has been introduced into remote sensing due to its ability to model long-range dependencies with linear complexity. This property makes it a promising alternative for efficient global spatial–spectral modeling. Representative Mamba-based methods include MambaHSI [42], which adaptively fuses spatial and spectral features, MambaLG [43], which integrates local and global spatial modeling with spectral awareness, and DBMGNet [44], which enhances spatial–spectral representation through a dual-branch Mamba-GCN framework.

Despite these advantages, Mamba-based methods still exhibit several limitations when applied to HSI classification tasks.

1) Despite their efficiency, existing Mamba-based methods still face two limitations in HSI classification. First, they mainly emphasize global contextual propagation, while local fine-grained variations such as boundaries, small targets, and class transition regions are not sufficiently modeled. A single sequence propagation pathway may weaken these high-frequency discriminative cues, making it difficult to jointly capture long-range dependencies and local details. Second, although hierarchical superpixels provide useful multiscale structural priors, existing methods usually rely on fixed cross-level mappings and shared transformations. Static mappings cannot adapt to feature distribution changes during training, while shared transformations ignore the scale-specific characteristics of pixel-level and superpixel-level features, leading to biased feature propagation and insufficient multiscale representation.
2) Hierarchical superpixels provide effective multiscale structural priors for HSI classification, which help alleviate pixel-level redundancy and enhance regional consistency modeling. However, existing methods usually rely on fixed cross-level mapping relationships and shared feature transformation strategies, making them inadequate for handling the pronounced differences among features at different scales. In particular, static mappings cannot be adaptively adjusted according to variations in feature distributions and may therefore introduce cross-level information propagation bias in complex scenes. Meanwhile, features at different scales have distinct modeling emphases, and a shared transformation strategy is often insufficient to capture their scale-specific characteristics, leading to multiscale feature mixing and weakened discriminative capability. Therefore, how to achieve adaptive optimization of cross-level mappings while preserving hierarchical structural priors, and how to perform differentiated modeling of multiscale features, remains a key challenge in hierarchical representation learning for HSIs.

Based on the above analysis, it can be seen that HSI classification simultaneously faces several challenges, including insufficient modeling of long-range spatial–spectral dependencies, inadequate characterization of local fine-grained variations, and ineffective collaboration among hierarchical multiscale features. To address these issues, this article proposes a hierarchical superpixel mixture-of-experts Mamba network for HSI classification, termed HSMMamba, from three aspects: hierarchical structural prior modeling, differentiated multiscale feature enhancement, and global–local joint representation learning. The proposed method aims to achieve adaptive optimization of cross-level mapping relationships, selective enhancement of multiscale features, and collaborative modeling of long-range dependencies and local detailed information within a unified framework, thereby improving spatial–spectral representation capability in complex scenes. The main contributions of this article are summarized as follows.

1) We propose a learnable hierarchical superpixel mapping (LHSM) mechanism to overcome the limitation of conventional hierarchical superpixel methods, in which fixed cross-level mapping relationships are difficult to adapt to complex land-cover structural variations. By introducing learnable residuals on top of hierarchical structural priors, the proposed mechanism adaptively optimizes cross-level mappings and dynamically adjusts the multiscale feature propagation process, thereby improving the consistency and accuracy between region-level representations and pixel-level reconstruction.
2) We design a hierarchical scale-aware mixture-of-experts enhancement mechanism to address the significant differences among features at different scales in terms of representation capability and semantic emphasis. By constructing a multibranch expert network for pixel-level, superpixel-level, and larger-scale regional features, the proposed mechanism performs targeted modeling of fine-grained boundary information, intermediate-scale structural transitions, and large-scale regional consistency, thereby enhancing the model's ability to characterize complex multiscale spatial structures and further improving classification performance and robustness.
3) We propose a forward–backward–differential joint modeling module, termed FBDMamba, to address the insufficient modeling of long-range spatial–spectral dependencies and the tendency to overlook local boundaries and fine-grained variations in HSIs. By constructing three collaborative branches, namely the forward branch, the backward branch, and the central

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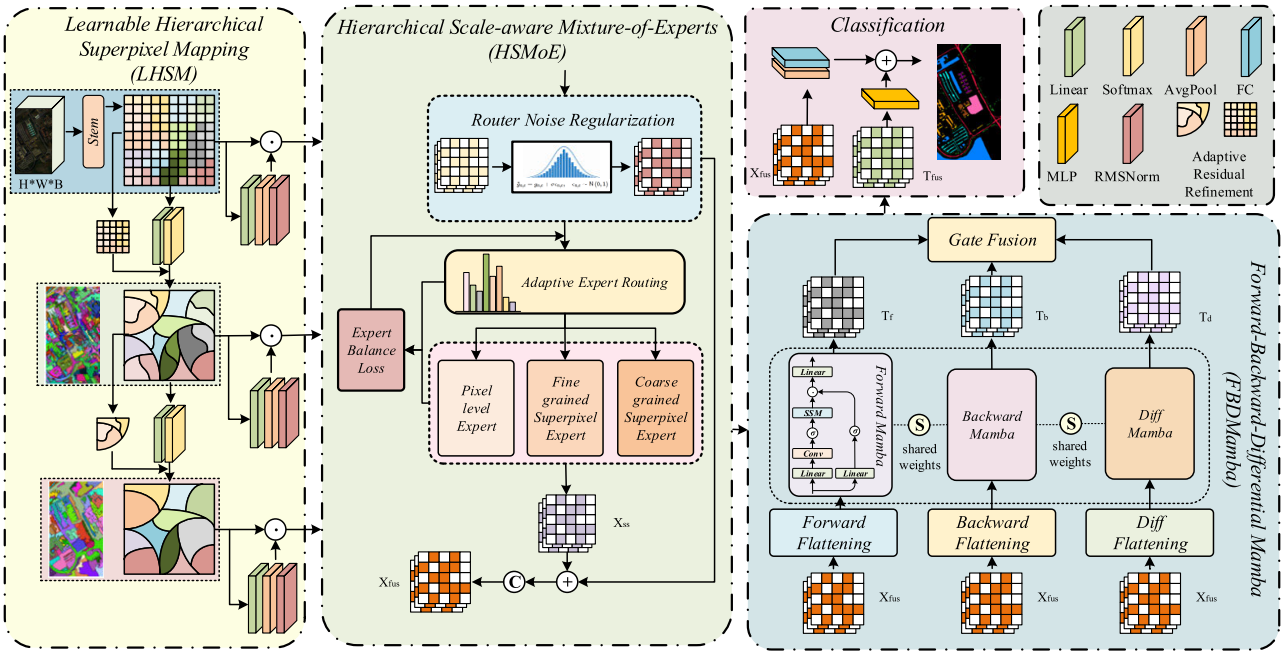


Fig. 1. Overall structural diagram of HSMMamba.

differential branch, the proposed module strengthens global contextual modeling while effectively capturing local variations and boundary structures, thereby significantly enhancing the discriminability and stability of feature representations in complex scenes.

4) We establish a unified HSMMamba framework that integrates hierarchical superpixel modeling, multiscale expert enhancement, and forward–backward–differential sequence modeling. Extensive experiments on Pavia University (PU), Salinas (SAs), and HongHu demonstrate that HSMMamba consistently outperforms mainstream HSI classification methods.

The remainder of this article is organized as follows. Section II presents the overall architecture of HSMMamba and details the operating principles of its individual modules. Section III reports extensive experimental results and comprehensive analyses, including ablation studies, quantitative comparisons, and visualization-based evaluations. Finally, Section IV concludes this article and discusses future research directions.

II. Methodology

We represent the input HSI as $X \in R^{H\times W\times B}$, where $H \times W$ represent the spatial dimensions of the input data and B represents the number of spectral bands in the HSIs. The overall architecture of HSMMamba is shown in Fig. 1. The input data are first processed by a stem module to obtain low-level features with reduced redundancy and unified dimensionality. Based on these features, HSMMamba progressively enhances spatial–spectral representations through hierarchical feature modeling. In particular, a LHSM module is introduced to adaptively refine cross-level mappings while preserving hierarchical structural priors, thereby improving the alignment and propagation of multiscale features. Then, a hierarchical scale-aware mixture-of-experts module performs differentiated enhancement of features at different scales, strengthening local details, regional consistency, and contextual information. Finally, the FBDMamba module jointly models features through forward, backward, and differential branches to capture long-range dependencies while emphasizing local boundaries and fine-grained variations. By integrating these components within a unified framework, HSMMamba provides robust feature representations for HSI classification, and the enhanced features from different levels are adaptively fused to generate the final prediction.

The three modules work collaboratively within a progressive framework. LHSM first constructs structurally aligned hierarchical features by refining the mappings between pixel-level and superpixel-level representations. HSMoE then adaptively enhances and fuses these multiscale features through scale-specific experts and a gating mechanism. Finally, FBDMamba further models the fused features by combining bidirectional long-range dependence modeling with differential local-variation enhancement, thereby improving both global contextual representation and local detail discrimination.

A. Learnable Hierarchical Superpixel Mapping

To reduce pixel-level redundancy and improve cross-scale information propagation, this article proposes a LHSM module. Although hierarchical superpixels provide useful multiscale structural priors, most existing methods rely on fixed cross-level mappings determined by the initial partition. Once the superpixel hierarchy is constructed, the pixel-to-region and region-to-pixel relationships remain unchanged during training, limiting their adaptability to dynamic feature distributions and complex land-cover structures. On the contrary, LHSM preserves the predefined hierarchical mapping as a structural prior while introducing learnable residual corrections to refine

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the mapping weights in a task-driven manner. This prior-guided learnable design enables adaptive cross-level feature propagation without discarding the stability of the original superpixel hierarchy.

LHSM first performs hierarchical partitioning on the feature map based on a predefined superpixel algorithm to construct a L-level superpixel structure. For the l level, the superpixel index to which pixel p belongs is denoted as $S^l(p) \in \{1, 2, \ldots n_l\}$, where n_l denotes the number of superpixel regions at that level. Different levels correspond to region partitions with different granularities, thereby forming a hierarchical representation from fine to coarse scales. The superpixel merging strategy used to construct the hierarchical partitions is illustrated in Fig. 2.

On this basis, feature aggregation is performed for each superpixel region. The regional feature of the j superpixel region at the l level is defined as

$$F_j^l = \frac{1}{|R_j^l|} \sum_{p \in R_j^l} F_p^0 \tag{1}$$

where $R_j^l = \{p | S^l(p) = j\}$ denotes the set of pixels contained in the jth superpixel at the lth level and $|R_j^l|$ is the number of elements in this set. Equation (1) maps pixel-level features to superpixel-level features through regional average aggregation, enabling the model to preserve local spectral details while strengthening the representation of intraregion consistency

$$S^{(p \to s_t)} = S^{fixed(p \to s_t)} + \Delta S^{(p \to s_l)} \tag{2}$$

where $S^{fixed(p \to s_t)}$ denotes the fixed mapping relationship obtained from superpixel partitioning and $\Delta S^{(p \to s_l)}$ denotes the learnable residual used to adaptively refine the cross-level mapping weights. Subsequently, rowwise Softmax normalization is applied to the mapping matrix to obtain

$$\tilde{S}_{i,:}^{(p \to s_t)} = softmax\left(S_{i,:}^{(p \to s_t)}\right). \tag{3}$$

Thereby ensuring that the weight distribution of each pixel during projection to the superpixel space satisfies the probability constraint. In this way, the pixel-level features are projected into the superpixel space through weighted aggregation and are then further transformed nonlinearly by a fully connected layer to obtain the output feature of the superpixels at the lth level

$$\hat{F}^l = \phi_l\left(\tilde{S}^{(p \to s_t)} F^0\right) = \sigma\left(W^l \tilde{S}^{(p \to s_t)} F^0 + b^l\right) \tag{4}$$

where W^l and b^l denote the weight matrix and bias term of the mapping function at the lth level, respectively and $\sigma(\cdot)$ denotes the nonlinear activation function. This design enables the cross-level mapping to go beyond static aggregation and to dynamically adjust the representation of regional features according to the task objective during training, thereby enhancing the discriminability and adaptability of region-level representations.

To enable region-level information to reversely supplement pixel-level representations, LHSM projects the transformed superpixel features back to the pixel positions. For an arbitrary pixel p, its mapped feature at the lth level is defined as

$$\tilde{F}_p^l = \hat{F}_{S^l(p)}^l \tag{5}$$

that is, pixel p inherits the enhanced representation of the superpixel region to which it belongs. Through this process, pixel-level features can establish stable connections with regional semantics at different levels, thereby preserving both local details and structural contextual information. Finally, the original pixel-level features are fused with the region-enhanced features obtained from each level to form the output representation

$$F^{\text{out}} = \psi\left(F^0, \tilde{F}^1, \ldots, \tilde{F}^L\right) \tag{6}$$

where $\psi(\cdot)$ denotes the feature fusion function, which can be implemented by channel concatenation followed by linear projection or by weighted fusion. As can be seen from (1) to (6), LHSM performs progressive modeling through "superpixel partitioning–regional aggregation–learnable mapping–pixel back-projection–multilevel fusion," thereby achieving adaptive optimization of cross-scale feature relationships while preserving hierarchical superpixel priors. The proposed module not only strengthens intraregion consistency and improves the accuracy of cross-level information propagation but also provides a more discriminative foundational representation for subsequent mixture-of-experts enhancement and sequence modeling.

B. HSMoE: Hierarchical Superpixel Mixture-of-Experts

After the hierarchical representations are constructed, the input features are organized into pixel-level and multiscale superpixel-level representations. Since these features differ significantly in spatial granularity and semantic emphasis, a shared transformation strategy is insufficient to fully exploit their complementarity. In particular, pixel-level features are more sensitive to local fine-grained spectral variations, whereas superpixel-level features are more suitable for describing regional consistency and contextual structures. Therefore, adaptive modeling and effective fusion of hierarchical heterogeneous features are crucial for improving representation capability.

Motivated by this observation, this article introduces a hierarchical superpixel mixture-of-experts mechanism to selectively enhance pixel-level and multiscale superpixel-level features. Instead of using a shared transformation for all hierarchical features, HSMoE constructs scale-specific expert branches to model their different characteristics: the pixel-level expert focuses on fine-grained spectral and boundary details, the intermediate-scale superpixel expert captures local structural transitions, and the large-scale superpixel expert emphasizes regional consistency and contextual information. A gating network is further employed to adaptively assign normalized weights to these experts according to the input features, thereby enabling dynamic fusion of heterogeneous multiscale representations. In this way, HSMoE better preserves scale-specific information and reduces feature mixing.

Across different levels, it improves the model's ability to characterize complex spatial structures and interclass differences. The detailed procedure is given in Algorithm 1.

Let the three hierarchical features obtained after LHSM be denoted as $X_p \in R^{N \times C}, X_{s1} \in R^{N \times C}, X_{s2} \in R^{N \times C}$, where X_p

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Algorithm 1 Process of HSMoE

Input: Hierarchical features X_p, X_{s1}, X_{s2}; expert index K = $\{p, s1.s2\}$

Output: Fused hierarchical representation X_{HSMoE} and balance loss L_{bal}

$\hat{X}_k \leftarrow Proj_k(X_k), \forall k \in K$
$E_k \leftarrow Expert_k(\hat{X}_k), \forall k \in K$
$g_k \leftarrow GAP(\hat{X}_k), \forall k \in K$
$g \leftarrow Concat(g_p, g_{s1}, g_{s2})$
$\alpha \leftarrow Softmax(Gate(g))$
$X_{HSMoE} \leftarrow \sum_{k \in K} \alpha_k \odot E_k$
$\bar{\alpha}_k \leftarrow Mean_{batch}(\alpha_k), \forall k \in K$
$L_{bal} \leftarrow \sum_{k \in K} \left(\bar{\alpha}_k - \frac{1}{K}\right)^2$
return X_{HSMoE}, L_{bal}

denotes the pixel-level feature and X_{s1} and X_{s2} denote the superpixel-level features at two different scales, respectively. Here, N denotes the number of feature units, and C denotes the feature dimension. To reduce the distribution discrepancy among representations at different hierarchical levels, they are first projected into a unified feature space through linear mappings, as follows:

$$\tilde{X}_p = W_p X_p, \tilde{X}_{s1} = W_{s1} X_{s1}, \tilde{X}_{s2} = W_{s2} X_{s2} \tag{7}$$

where W_p, W_{s1}, and W_{s2} are learnable projection parameters and $\tilde{X}_p, \tilde{X}_{s1}$, and $\tilde{X}_{s2}$ denote the aligned hierarchical feature representations. Subsequently, HSMoE treats the three hierarchical branches as three experts with different functional pReferences and performs independent transformations on them. The corresponding expert outputs can be written as follows:

$$E_p = f_p\left(\tilde{X}_p\right), E_{s1} = f_{s1}\left(\tilde{X}_{s1}\right), E_{s2} = f_{s2}\left(\tilde{X}_{s2}\right) \tag{8}$$

where $f_p(\cdot), f_{s1}(\cdot), f_{s2}(\cdot)$ denotes the nonlinear mapping functions corresponding to the pixel-level expert, the intermediate-scale superpixel expert, and the large-scale superpixel expert, respectively. Through independent expert modeling, features at different hierarchical levels can be enhanced in more suitable representation spaces, thereby avoiding the feature confusion caused by a shared transformation strategy.

Before adaptive fusion, the three hierarchical representations are independently processed by their corresponding expert branches to preserve scale-specific characteristics. The pixel-level expert is mainly responsible for enhancing fine-grained spectral responses and boundary-sensitive details, which are important for distinguishing categories with subtle spectral differences. The intermediate-scale superpixel expert focuses on local structural transitions and neighborhood consistency, helping the model capture spatial patterns between pixel-level details and regional semantics. The large-scale superpixel expert further emphasizes broader contextual information and regional homogeneity, which is beneficial for suppressing isolated noise and improving semantic consistency. Through such scale-specific expert modeling, hierarchical features are enhanced in more suitable representation spaces before being adaptively fused.

To achieve adaptive selection and effective collaboration among hierarchical features, HSMoE introduces a gating network to estimate the contributions of the three expert branches. In particular, global descriptors are extracted from the pixel-level, intermediate-scale superpixel-level, and large-scale superpixel-level features, and are jointly fed into the gating network to generate normalized weights for the corresponding experts. These weights characterize the relative importance of different hierarchical branches under a unified constraint. Based on the learned weights, the expert outputs are adaptively fused to produce the final HSMoE representation. Unlike fixed fusion strategies, this mechanism dynamically adjusts the contribution of each branch according to the input scene and feature distribution, thereby enhancing the complementarity of heterogeneous multiscale features.

It should be noted that relying solely on the gating network for dynamic allocation may lead to imbalanced expert utilization during training, where some experts dominate for a long period while others are insufficiently trained, thereby limiting the representation capability of the model. To address this issue, this article further introduces an expert balance loss to constrain the utilization of the three experts and encourage the gating network to distribute activation probabilities more evenly among them. Within a batch, the average gating weights of the three experts are given by

$$\tilde{\alpha}_p = \frac{1}{N}\sum_{i=1}^{N} \alpha_p^{(i)}, \tilde{\alpha}_{s1} = \frac{1}{N}\sum_{i=1}^{N} \alpha_{s1}^{(i)}, \tilde{\alpha}_{s2} = \frac{1}{N}\sum_{i=1}^{N} \alpha_{s2}^{(i)} \tag{9}$$

where $\alpha_p^{(i)}, \alpha_{s1}^{(i)}, \alpha_{s2}^{(i)}$ denotes the gating weight corresponding to the ith feature unit. To prevent the gating distribution from being overly biased toward a particular expert, the balancing objective is defined to be close to a uniform distribution, and the following expert balance loss is constructed:

$$L_{bal} = \sum_{j \in \{p, s1, s2\}} \left(\bar{\alpha}_j - \frac{1}{3}\right)^2. \tag{10}$$

The expert balance loss is not merely an auxiliary constraint introduced to prevent an expert from becoming inactive; rather, it suppresses, from an optimization perspective, the single-expert dominance phenomenon caused by data distribution bias or gradient competition during the early stage of training. By explicitly constraining the gating distribution, the model enables different experts to obtain more sufficient and balanced updates during training, thereby improving the adequacy of collaborative expert learning and the overall training stability. In other words, this balancing mechanism makes HSMoE more than a simple dynamic routing structure, turning it into a complete feature enhancement mechanism that jointly accounts for adaptive selection and stable collaborative learning

$$L = L_{cls} + \lambda L_{bal} \tag{11}$$

where L_{cls} denotes the main classification loss and λ is the balancing coefficient.

In summary, HSMoE transforms multiscale heterogeneous feature modeling into a unified process of expert specialization, adaptive selection, and collaborative regularization.

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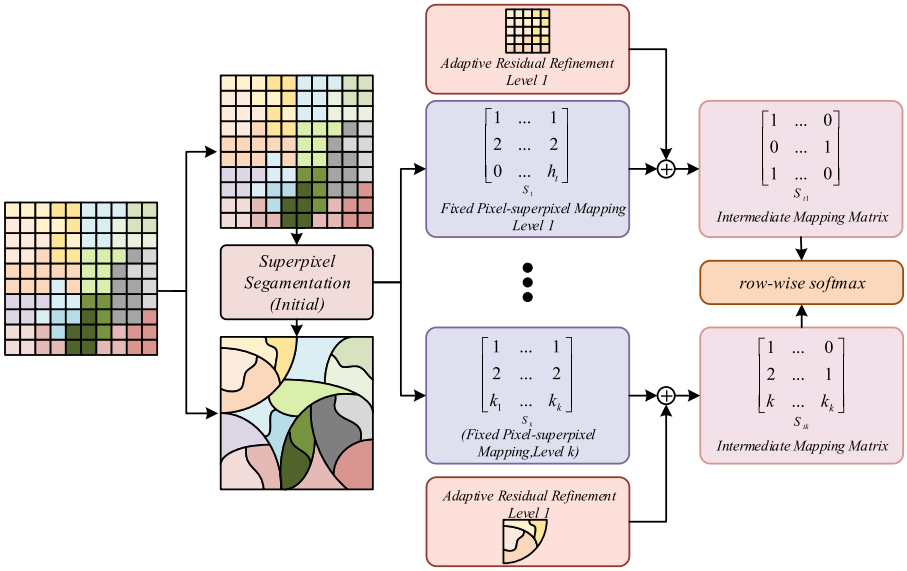


Fig. 2. Illustration of the superpixel merging strategy.

By doing so, it provides more robust and discriminative hierarchical fused features for subsequent sequence modeling.

C. FBDMamba: Forward–Backward–Differential Mamba

After hierarchical feature enhancement and fusion, the input representation contains both local detailed information and multiscale regional structural information, thereby providing a more robust basis for subsequent modeling. However, although Mamba exhibits a clear advantage in modeling long-range dependencies and can effectively capture global spatial–spectral contextual relationships in HSIs, its ability to characterize local fine-grained variations, boundary details, and class transition regions remains limited. This is because the recursive propagation mechanism of the state-space model is inherently more suitable for describing continuous and smooth contextual dependencies, whereas boundaries, small-scale targets, and transition regions in HSIs usually appear as locally discontinuous or abruptly varying patterns. If only global contextual propagation is considered, such local high-frequency discriminative cues can be easily weakened, which in turn degrades classification performance in complex scenes.

To further illustrate this issue, Fig. 3 presents a comparison between the real image of the PU dataset and its corresponding differential image. Compared with the original image, the differential representation highlights regional boundaries, texture variations, and class transition locations more clearly while suppressing part of the redundant responses in smooth regions. This observation indicates that differential information is naturally effective for characterizing local structural variations and high-frequency discriminative cues. Motivated by this, a differential branch is further introduced on top of bidirectional Mamba modeling to explicitly enhance the representation of local variation patterns.

Based on the above considerations, this article proposes a forward–backward–differential Mamba module, termed FBDMamba, to preserve the strength of Mamba in global



(a) (b)

Fig. 3. Visualization comparison of the PU dataset. (a) RGB image. (b) Differential image.

dependence modeling while enhancing the characterization of local details and boundary variations. The overall structure is shown in Fig. 4. Through the collaborative design of forward modeling, backward complementation, and differential enhancement, the proposed module achieves joint representation of global contextual information and local variation information within a unified framework.

In particular, the forward branch captures global spatial dependencies along the image sequence, while the backward branch models complementary dependencies in the reverse direction. This bidirectional design enables FBDMamba to capture long-range correlations from complementary propagation directions and reduces the information bias caused by unidirectional state propagation. On the contrary, the differential branch focuses on the discrepancy between the forward and backward representations, thereby highlighting local discontinuities, boundary variations, and fine-grained texture changes. Therefore, the differential branch complements bidirectional contextual modeling by explicitly strengthening high-frequency discriminative cues, such as class transitions and boundary contours.

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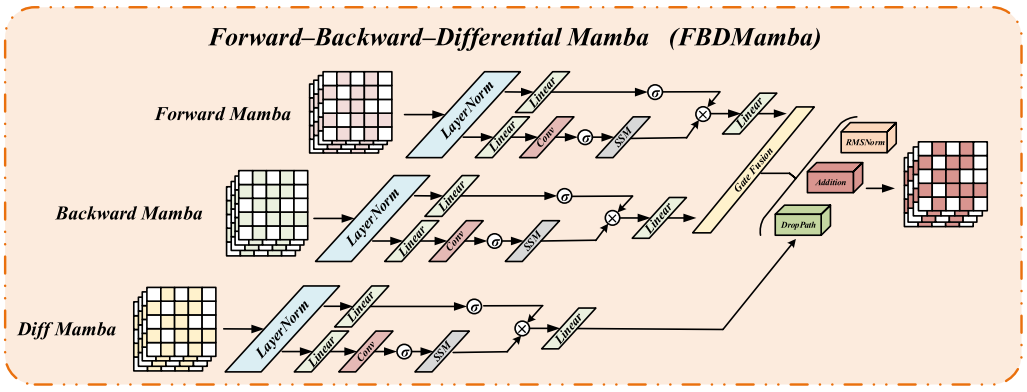


Fig. 4. Structural diagram of FBDMamba.

To integrate these complementary representations, a gated fusion mechanism is adopted to adaptively regulate the contribution of the forward, backward, and differential branches. The gating network generates branch-specific weights according to the input feature distribution, enabling flexible fusion of global dependence information and local differential cues. Compared with fixed-weight fusion strategies, this adaptive fusion mechanism better balances contextual consistency and local detail discrimination under different scene conditions.

To further improve training stability and robustness, a residual connection is introduced after fusion, together with DropPath and RMSNorm. The residual connection preserves the original input information and alleviates feature degradation, DropPath mitigates overfitting by randomly dropping paths during training, and RMSNorm stabilizes feature normalization. Through this design, FBDMamba effectively unifies bidirectional long-range dependence modeling with differential local-variation enhancement, providing more discriminative representations for complex hyperspectral scenes.

III. Experimental Results and Analysis

To comprehensively evaluate the performance of the proposed HSMMamba, experiments were conducted on three representative hyperspectral datasets, namely PU, SAs, and HongHu acquired by Wuhan University. These datasets exhibit significant differences in scene characteristics and spectral properties, making them suitable for assessing the generalization capability and classification effectiveness of the proposed method from multiple perspectives. Detailed descriptions of the datasets are provided as follows.

A. Dataset Description

To ensure a fair comparison, the same sample partition strategy was adopted for all datasets. In particular, for each dataset, 30 labeled samples from each class were randomly selected for training, and ten labeled samples from each class were used for validation. The remaining labeled samples were used for testing. This setting was adopted to evaluate the classification performance of different methods under limited training samples.

PU: The PU dataset was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor over Pavia, Italy. It has a spatial resolution of 1.3 m and an image size of 610 × 340 pixels, with 103 spectral bands. The ground-truth map covers nine land-cover categories, including trees, bare soil, and asphalt roads.

SAs: The SAs dataset was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the SAs Valley, California, USA. The image size is 512 × 217 pixels with a spatial resolution of 3.7 m, and the dataset contains 204 spectral channels. It includes 16 land-cover categories, covering various crop types such as grapes, lettuce, and vineyard parcels.

HongHu: The HongHu dataset was collected in HongHu, Hubei Province, China, from 16:23 to 17:37 on November 20, 2017, using a Headwall Nano-Hyperspec hyperspectral imager (focal length: 17 mm) mounted on a DJI Matrice 600 Pro UAV platform. This area represents a typical complex agricultural scene, containing not only multiple crop types but also several varieties of the same crop. The image size is 940 × 475 pixels, with a spectral range of 400–1000 nm and 270 spectral channels.

B. Experimental Setup

All experiments were conducted using PyTorch on a platform equipped with an Intel[1] Core[2] i9-14900KF CPU, an NVIDIA GeForce RTX 4090D GPU, and 128 GB of RAM. Classification performance was evaluated using three widely adopted metrics, namely overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

To validate the effectiveness of HSMMamba, several representative HSI classification methods were selected for comparison, including SpectralFormer [20], CLOLN [21], MorphFormer [23], SSFTT [24], GSCViT [25], MambaHSI [42], MambaLG [43], and DBMGNet [44]. These methods cover advanced architectures based on Transformers, CNNs, GCNs, and Mamba, enabling a comprehensive evaluation of the proposed model. Among them, SpectralFormer and SSFTT enhance spectral–spatial representation through Transformer-based sequence modeling, CLOLN emphasizes lightweight spectral–spatial learning, MorphFormer and GSCViT improve

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[2]Trademarked.

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TABLE I
IMPACT OF DIFFERENT MODULES ON OA FOR THREE DATASETS

Case	Component				DataSets		
	Baseline	1	2	3	PU	SA	HH
1	√	—	—	—	93.65	94.49	90.48
2	√	√	—	—	95.17	96.02	92.54
3	√	√	√	—	97.18	98.71	93.24
4	√	√	—	√	96.41	98.20	94.27
5	√	√	√	√	**98.98**	**99.30**	**95.65**

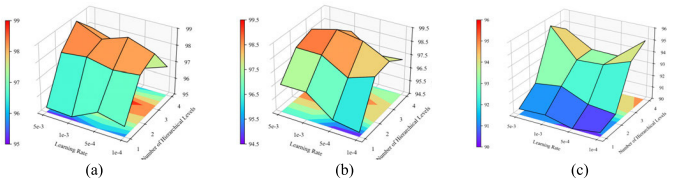


Fig. 5. Effect of different hierarchical levels and learning rates on OA across different datasets. (a) PU dataset. (b) SA dataset. (c) HongHu dataset.

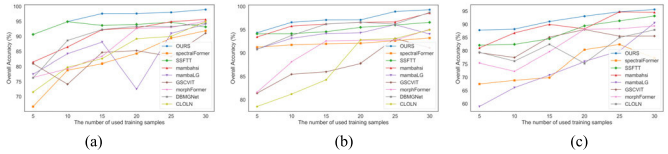


Fig. 6. Effect of different numbers of training samples on OA for different methods. (a) PU dataset. (b) SA dataset. (c) HongHu dataset.

joint local–global modeling, while MambaHSI, MambaLG, and DBMGNet further explore efficient long-range dependence modeling with Mamba-based architectures.

C. Ablation Experiment

To evaluate the effectiveness of the proposed components, ablation experiments were conducted on the PU, SA, and HH datasets. Starting from a baseline model, LHSM, HSMoE, and FBDMamba were progressively introduced to construct five configurations, and the results are reported in Table I. In particular, Case 1 denotes the baseline, Case 2 adds LHSM, Case 3 further introduces HSMoE, Case 4 adds FBDMamba on top of Case 2, and Case 5 corresponds to the complete model.

As shown in Table I, the classification performance consistently improves as the proposed modules are progressively incorporated, demonstrating the effectiveness of each component. Compared with the baseline, introducing LHSM improves the performance by 1.52%, 1.53%, and 2.06%

On the PU, SA, and HH datasets, respectively, showing its ability to enhance cross-level feature propagation and regional structural representation. Adding HSMoE further improves the accuracy by 2.01%, 2.69%, and 0.70%, indicating its effectiveness in adaptively enhancing features at different hierarchical levels. After incorporating FBDMamba, the accuracies reach 96.41%, 98.20%, and 94.27%, further confirming its advantage in jointly modeling global dependencies and local details. Finally, the complete model achieves gains of 5.33%, 4.81%, and 5.17% over the baseline on the three datasets, respectively, which demonstrates the strong complementarity among LHSM, HSMoE, and FBDMamba.

D. Parameter Sensitivity Analysis

To further analyze the sensitivity of HSMMamba to key hyperparameters, experiments were conducted on the PU, SA, and HH datasets to evaluate the OA under different learning rates and different settings of the number of hierarchical levels, as shown in Fig. 5. The results indicate that all three datasets achieve the best or near-best performance when the number of hierarchical levels is set to 3, suggesting that a three-level hierarchy provides a favorable balance between hierarchical information richness and feature propagation stability. When the number of levels is too small, the model cannot fully exploit multilevel superpixel priors, whereas an overly deep hierarchy introduces redundant information and increases the difficulty of cross-level mapping.

The model also shows noticeable sensitivity to the learning rate. In general, excessively large learning rates tend to cause unstable convergence, while overly small learning rates reduce optimization efficiency. By comparison, learning rates around $1e^{-3}$ and $5e^{-4}$ usually yield better and more stable performance. Therefore, the number of hierarchical levels is fixed to 3 in the subsequent experiments.

TABLE II
COMPARISON OF DIFFERENT FEATURE FUSION STRATEGIES ON THE PU, SA, AND HONGHU DATASETS

Methods	OA		
	PU	SA	HH
Add	95.74	97.97	94.36
Concat	96.30	98.27	94.82
Gate	96.15	98.42	94.99
BSConv	96.59	98.63	94.60
HSMoE	98.98	99.30	95.65

E. Comparison of Different Training Sample Sizes

To evaluate the classification capability of HSMMamba under different training-sample settings, different numbers of training samples were specified on the PU, SA, and HH

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TABLE III
QUANTITATIVE RESULT (ACC% ± STD%) OF THE PU DATASET

Methods	SpectralFormer TGRS 2021	CLOLN TGRS2024	MorphFormer TGRS2023	SSFTT TGRS2022	GSCViT TGRS2024	MambaHSI TGRS2024	MambaLG TGRS2025	DBMGNet TGRS2025	HSMMamba Ours
1	85.92±2.36	98.27±1.00	95.72±0.79	90.27±1.28	97.10±1.64	94.35±1.73	93.96±2.36	95.59±0.15	**98.89±1.37**
2	92.97±2.39	98.98±0.76	95.46±1.62	94.58±4.84	97.68±1.32	96.43±1.73	93.57±0.94	**99.21±0.78**	99.04±0.36
3	84.73±5.17	87.83±7.54	93.52±0.34	90.41±4.20	86.46±7.42	93.39±5.35	89.95±1.55	**74.73±0.51**	**99.61±0.02**
4	**98.90±5.24**	97.80±1.46	97.13±2.49	91.67±1.36	91.83±9.94	89.75±3.60	97.10±1.17	88.09±1.66	96.2±0.28
5	99.92±1.88	99.68±0.59	99.84±0.87	99.71±0.20	99.45±0.53	99.00±0.02	99.99±0.01	99.98±0.81	**100.00±0**
6	88.40±9.24	85.27±7.50	89.95±0.91	95.99±2.07	79.65±9.24	98.60±0.93	**99.28±0.18**	92.10±3.43	98.87±1.07
7	93.85±2.06	85.92±7.00	**99.53±0.61**	98.26±1.06	86.63±10.10	96.50±3.13	99.25±0.75	71.53±13.42	97.64±1.25
8	95.20±5.28	90.93±4.12	93.04±0.69	88.33±3.29	89.50±4.70	94.40±2.93	83.84±2.82	92.16±2.88	**99.94±0.22**
9	98.98±0.71	97.12±2.03	99.12±0.97	96.31±2.05	98.71±1.10	99.36±0.62	99.89±0.94	97.06±4.67	**99.95±0.01**
OA(%)	91.89±3.25	94.97±1.43	95.08±1.53	93.17±2.41	91.10±2.60	95.74±0.90	94.07±0.38	94.32±0.88	**98.98±0.28**
AA(%)	93.21±1.07	93.53±1.59	95.29±1.49	93.51±0.68	83.93±2.67	95.86±1.11	95.28±0.76	90.05±0.02	**98.90±1.00**
Kappa×100	89.24±3.19	93.38±1.84	93.40±0.72	90.87±2.59	77.40±4.41	90.38±2.95	92.23±3.62	92.42±0.11	**98.65±0.81**

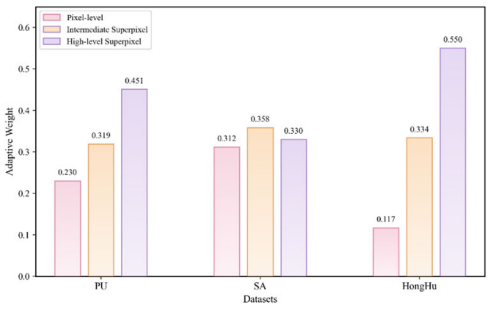


Fig. 7. Adaptive weight distributions of different hierarchical branches on the PU, SA, and HongHu datasets.

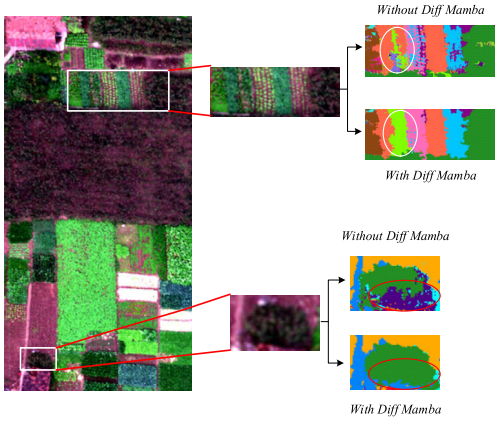


Fig. 8. Visualization comparison of classification results with and without the differential Mamba branch on the HongHu dataset.

datasets, and the proposed method was compared with multiple competing approaches. The corresponding results are shown in Fig. 6. For all three datasets, the number of training samples per class was set to {5, 10, 15, 20, 25, 30}, and the resulting OA was averaged over ten runs with different random seeds. In particular, Fig. 6(a)–(c) illustrates the effect of the number of training samples on OA for the PU, SA, and HongHu datasets, respectively. It can be observed that, as the number of training samples increases, the classification accuracy of all methods generally improves. Nevertheless, HSMMamba consistently achieves superior overall performance on all three datasets, demonstrating favorable stability and robustness.

F. Analysis of Adaptive Weights Across Different Hierarchical Branches

To further analyze the adaptive behavior of HSMoE, the weight distributions of different hierarchical branches are visualized in Fig. 7 for the PU, SA, and HongHu datasets. The weights vary across datasets, indicating that HSMMamba can adjust the contributions of pixel-level, intermediate-scale superpixel-level, and high-level superpixel-level features according to different scene characteristics. In particular, the pixel-level branch mainly captures fine-grained spectral and boundary details, the intermediate-scale branch focuses on local structural transitions, and the high-level branch emphasizes regional consistency and contextual information.

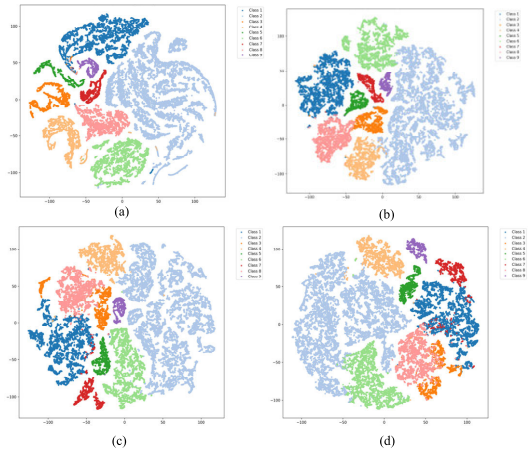


Fig. 9. t-SNE visualization results on the PU dataset. (a) HSMMamba. (b) GSCViT. (c) MorphFormer. (d) SSFTT.

As shown in Fig. 7, PU and HongHu assign relatively higher weights to high-level superpixel features, suggesting that regional context is important for urban scenes with complex spatial layouts and agricultural scenes with fine-grained categories. On the contrary, SA shows a more balanced weight distribution, indicating that both

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TABLE IV
QUANTITATIVE RESULT (ACC% ± STD%) OF SA DATASET

Methods	SpectralFormer TGRS 2021	CLOLN TGRS2024	MorphFormer TGRS2023	SSFTT TGRS2022	GSCViT TGRS2024	MambaHSI TGRS2024	MambaLG TGRS2025	DBMGNet TGRS2025	HSMMamba Ours
1	97.42±1.57	99.98±2.54	98.43±2.35	99.95±0.11	99.91±0.08	99.93±0.18	100±0	99.30±0.31	**100±0**
2	98.82±0.62	100±0.02	99.16±1.25	99.88±0.28	100±0.01	100±0.01	99.92±1.28	99.62±0.83	**100±0**
3	97.43±1.48	96.51±2.35	99.94±0.86	100±0	99.93±0.18	99.92±0.15	99.94±0.14	99.81±1.41	**99.99±0.02**
4	99.50±0.15	99.64±1.31	99.88±0.04	99.89±0.07	**99.95±0.11**	99.90±0.09	99.25±0.58	99.72±1.24	99.84±0.44
5	95.77±2.38	98.97±0.06	96.90±0.33	97.95±1.70	98.20±0.53	98.25±0.92	98.83±0.66	**99.99±0.01**	98.22±0.93
6	98.98±0.39	99.98±0.02	100±0.01	99.96±0.06	99.99±0.01	99.99±0.03	**99.99±0.02**	97.50±0.42	99.94±0.04
7	99.43±0.11	99.70±1.23	96.84±0.41	99.95±0.11	99.80±0.29	99.80±0.24	99.97±0.03	**99.99±0.01**	98.22±0.93
8	71.16±7.84	97.55±0.24	85.98±1.03	90.41±3.35	97.28±1.72	96.08±2.35	87.78±1.25	97.50±0.42	**99.90±0.08**
9	97.33±1.29	99.96±1.05	99.93±0.16	99.92±0.14	99.97±1.65	99.97±0.07	99.96±0.02	99.41±0.51	**99.99±0.03**
10	88.12±5.60	97.60±0.58	96.45±0.96	98.03±1.23	97.69±0.38	97.55±1.54	96.34±0.29	97.93±0.97	**98.45±1.24**
11	96.88±0.89	99.68±0.52	99.32±0.56	**99.96±0.08**	99.71±0.37	99.91±0.09	99.33±0.92	98.34±1.80	99.27±1.00
12	99.05±0.99	97.25±0.28	99.98±0.07	99.88±0.20	99.87±0.09	99.99±0.03	99.99±0.04	98.51±0.96	**99.99±0.01**
13	99.19±0.34	99.24±1.00	99.96±0.02	99.65±0.57	99.88±0.08	99.75±0.51	**99.97±0.03**	99.86±0.43	99.50±0.59
14	95.37±3.03	99.12±0.67	97.40±0.32	95.35±7.32	99.89±0.25	99.88±0.20	**100±0**	99.47±0.25	98.55±0.38
15	67.02±5.92	98.88±2.01	86.81±0.95	92.31±4.17	98.23±1.24	97.76±1.85	94.14±0.72	96.00±1.39	**98.57±0.26**
16	95.81±2.30	99.01±0.24	99.32±0.56	98.95±1.02	99.90±0.10	99.90±0.14	99.44±0.63	99.26±0.79	**99.92±0.16**
OA(%)	93.23±1.76	98.68±0.77	94.50±0.53	96.58±0.95	98.23±0.41	98.60±0.43	96.31±0.87	98.72±0.26	**99.30±0.35**
AA(%)	95.99±2.20	98.17±0.14	97.20±0.59	98.25±0.84	99.17±0.30	99.24±0.20	96.31±0.87	99.23±0.20	**99.39±0.23**
Kappa×100	92.46±1.96	99.18±0.38	93.88±0.42	96.20±1.87	99.17±0.30	99.00±0.95	98.44±0.33	99.23±0.29	**99.32±0.10**

TABLE V
QUANTITATIVE RESULT (ACC% ± STD%) OF HONGHU DATASET

Methods	SpectralFormer TGRS 2021	CLOLN TGRS2024	MorphFormer TGRS2023	SSFTT TGRS2022	GSCViT TGRS2024	MambaHSI TGRS2024	MambaLG TGRS2025	DBMGNet TGRS2025	HSMMamba Ours
1	97.50±1.26	95.36±5.56	92.64±0.86	94.66±1.33	**98.16±1.08**	96.47±1.50	92.91±0.68	96.43±0.96	96.25±2.49
2	94.99±4.37	78.60±6.19	92.21±1.16	91.09±0.70	86.90±9.09	95.43±2.33	90.81±3.69	91.74±3.87	**96.98±2.43**
3	86.09±4.51	**98.22±1.32**	81.32±1.53	90.69±3.40	79.45±1.81	94.66±1.05	84.59±0.94	89.79±6.90	95.31±3.17
4	94.18±3.57	99.48±0.37	97.09±2.78	93.10±2.81	99.57±0.14	96.96±1.79	96.20±1.51	**99.87±0.93**	97.99±3.91
5	96.68±2.78	41.33±18.92	92.77±1.31	87.13±0.63	37.03±9.33	96.20±1.77	93.47±2.21	68.24±2.64	**97.41±1.57**
6	92.24±1.93	96.09±1.67	91.81±3.95	94.20±0.82	95.72±1.83	93.90±2.47	94.07±0.69	97.72±0.51	**97.95±4.39**
7	81.36±3.21	**92.72±1.96**	64.15±4.86	73.01±0.95	90.38±3.66	86.60±3.30	72.65±2.09	91.03±0.95	88.35±4.56
8	80.30±2.31	46.44±6.02	69.28±4.29	93.58±4.69	32.12±4.83	96.08±2.90	77.98±0.87	43.07±2.40	**99.67±2.17**
9	97.64±1.71	97.04±2.05	95.30±0.76	99.02±2.67	97.06±2.73	96.08±2.91	93.65±1.35	95.55±0.99	**99.07±0.46**
10	74.88±7.08	84.88±3.47	82.34±3.90	83.46±0.81	79.23±7.53	90.45±2.72	89.39±2.68	90.17±1.85	**94.66±2.78**
11	83.33±5.48	72.03±8.35	73.45±1.09	79.97±5.34	72.41±8.95	91.96±2.87	77.11±7.80	83.90±4.75	**92.74±3.93**
12	82.78±4.37	66.63±8.82	76.22±1.42	78.80±0.94	65.61±8.66	87.40±5.22	79.92±5.05	82.46±0.90	**94.38±3.17**
13	79.48±4.52	82.67±5.83	71.82±4.53	**94.14±1.44**	77.49±8.47	89.12±3.12	68.96±4.63	87.42±0.01	91.59±1.93
14	91.91±6.77	87.59±7.03	93.50±2.59	94.25±1.25	86.26±11.93	94.01±1.65	88.92±4.31	94.50±0.78	**94.88±0.86**
15	97.62±4.25	75.88±12.93	**99.48±0.59**	93.92±5.97	63.62±20.25	98.84±1.17	97.33±1.49	88.18±5.14	98.33±0.29
16	98.20±2.55	97.39±1.73	97.37±2.78	98.95±0.97	97.42±1.66	96.83±1.60	93.56±8.44	**99.24±0.06**	94.48±1.83
17	92.32±4.87	80.09±9.70	91.87±0.91	99.46±1.58	84.92±6.09	98.60±1.49	95.17±7.11	65.94±3.43	**99.55±2.65**
18	98.40±3.59	65.55±15.89	91.05±2.07	87.94±0.56	67.44±10.50	96.72±1.82	**98.71±1.89**	92.64±1.34	97.24±0.93
19	92.10±5.12	82.53±7.23	89.50±0.70	85.59±0.27	85.10±8.11	91.30±3.07	91.51±2.57	**95.07±2.88**	89.66±5.83
20	97.17±5.53	81.38±6.60	96.99±0.74	94.05±1.51	71.64±10.46	**99.18±0.42**	92.62±3.14	94.76±0.46	98.25±3.57
21	97.77±0.92	30.19±12.06	81.20±1.84	100±0	25.92±7.91	99.91±0.18	98.61±3.67	40.21±4.55	**100±0**
22	98.32±1.27	55.02±10.89	92.16±0.95	96.38±4.03	46.26±11.15	**99.54±0.51**	97.86±3.68	79.08±1.98	98.78±0.94
OA(%)	90.87±1.27	87.93±2.11	89.40±3.49	91.97±2.44	85.55±1.86	94.58±1.01	90.56±3.60	94.25±0.72	**95.65±1.59**
AA(%)	91.15±0.44	77.60±2.25	86.98±3.03	91.20±0.61	83.21±1.64	94.77±0.61	89.71±3.00	85.05±1.48	**96.07±1.91**
Kappa×100	88.58±1.28	85.06±2.49	86.71±2.41	89.94±2.55	82.07±2.16	94.00±1.88	88.17±7.11	92.70±1.54	**94.51±1.89**

spectral details and regional structures contribute to classification in crop-dominated scenes. These results demonstrate that the proposed HSMoE can adaptively exploit hierarchical information under different scene conditions, thereby improving multiscale feature fusion and classification robustness.

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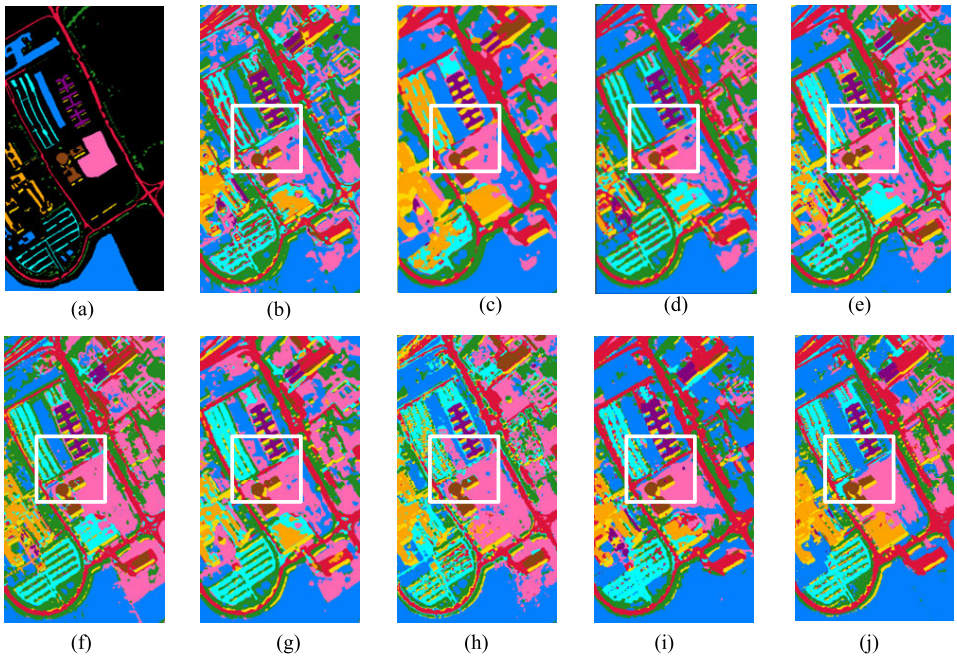


Fig. 10. Classification maps obtained by different methods on the PU dataset. (a) Ground truth map. (b) SpectralFormer. (c) CLOLN. (d) DBMGNet. (e) MorphFormer. (f) SSFTT. (g) GSCViT. (h) MambaLG. (i) MambaHSI. (j) Proposed HSMMamba.

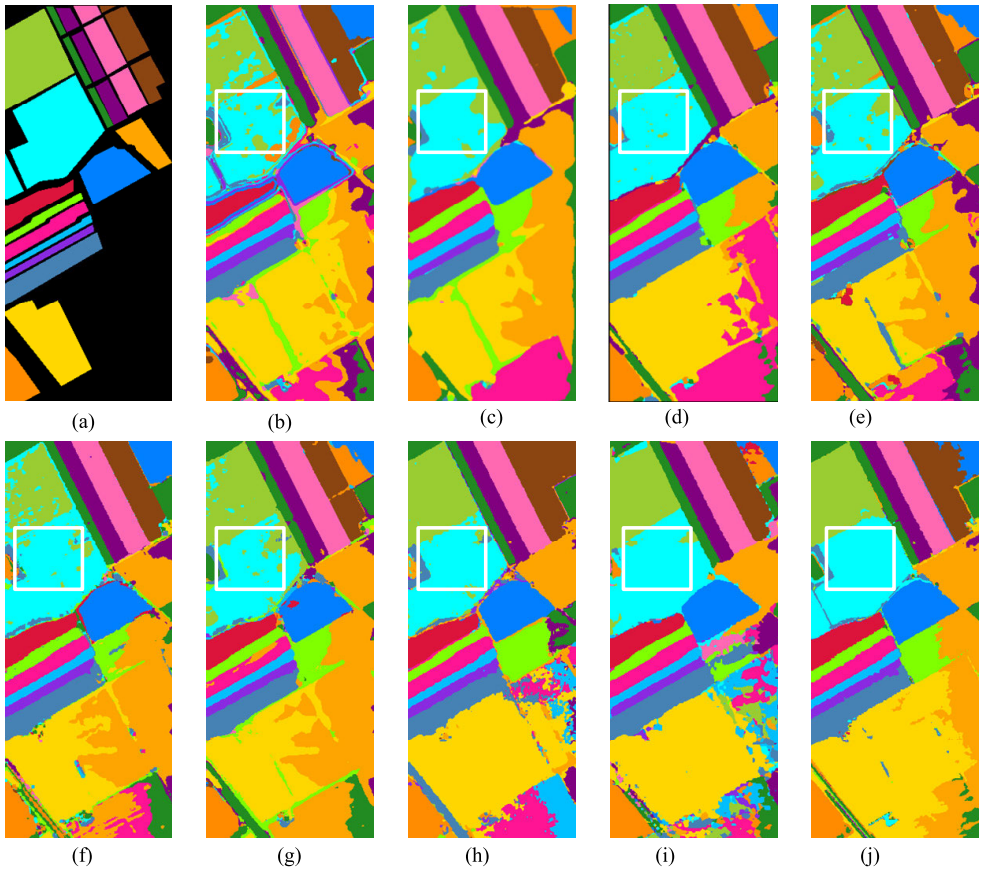


Fig. 11. Classification maps obtained by different methods on the SA dataset. (a) Ground truth map. (b) SpectralFormer. (c) CLOLN. (d) DBMGNet. (e) MorphFormer. (f) SSFTT. (g) GSCViT. (h) MambaLG. (i) MambaHSI. (j) Proposed HSMMamba.

G. Effectiveness Analysis of the Differential Mamba Branch

To further validate the effectiveness of the differential Mamba branch (DiffMamba) in FBDMamba for HSI classification, a comparative analysis of local prediction results was conducted on the HongHu dataset between the bidirectional Mamba model with the differential branch and its counterpart without the differential branch, namely FBMamba.

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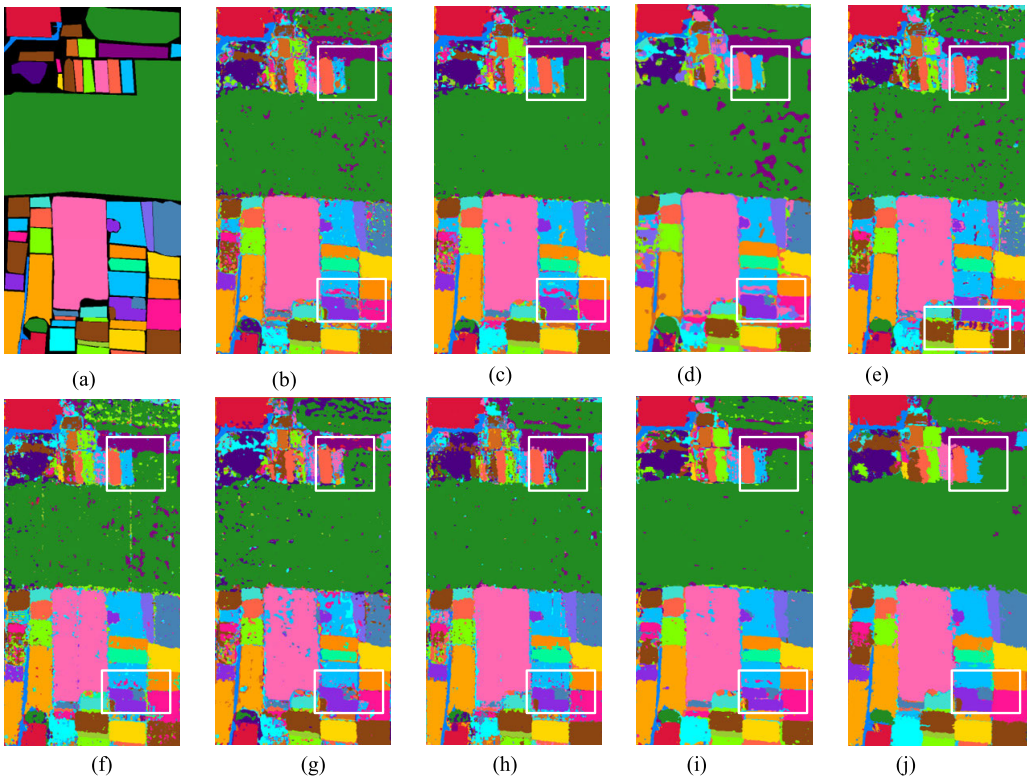


Fig. 12. Classification maps obtained by different methods on the HongHu dataset. (a) Ground truth map. (b) SpectralFormer. (c) CLOLN. (d) DBMGNet. (e) MorphFormer. (f) SSFTT. (g) GSCViT. (h) MambaLG. (i) MambaHSI. (j) Proposed HSMMamba.

The visualization comparison between the models with and without the differential Mamba branch is shown in Fig. 8.

In particular, two representative local regions were selected from the overall classification maps and enlarged for visualization, in order to examine the differences in model performance on fine-grained boundaries, local structural variations, and class transition regions. The visualization results show that, after introducing DiffMamba, the predictions in local regions become more continuous and refined, with clearer boundary contours and significantly reduced misclassification in small structures and transition areas. On the contrary, the model using only bidirectional Mamba still exhibits blurred boundaries and local discontinuities in some complex regions. These results demonstrate that, by introducing a central differential branch, DiffMamba effectively enhances the perception of local fine-grained variations, thereby further improving classification accuracy and robustness in complex scenes.

H. Effectiveness of HSMoE in Multiscale Feature Fusion

To evaluate the effectiveness of HSMoE in multiscale feature fusion, this article compares it with several common feature fusion strategies, including Gate, Add, Concat, and BSConv. During the experiments, all other network components were kept unchanged, and only the feature fusion module was replaced to ensure a fair comparison. All methods were evaluated on the PU, SAs, and HongHu datasets, with OA adopted as the evaluation metric.

The experimental results are reported in Table II. It can be observed that, compared with several conventional fusion strategies, HSMoE achieves superior OA performance on all three datasets, demonstrating stronger discriminative capability and adaptability in the feature fusion process. Compared with static fusion strategies such as Gate, Add, and Concat, HSMoE can perform targeted enhancement of features at different scales through the mixture-of-experts mechanism, thereby more effectively preserving fine-grained boundary information and regional structural information. Compared with BSConv, HSMoE exhibits greater flexibility and robustness in handling heterogeneous features across different scales. These results indicate that the proposed HSMoE can more fully exploit the complementary advantages of multiscale features, thereby improving HSI classification performance.

I. T-SNE Visualization Results and Analysis

To more intuitively evaluate the discriminative capability of the features learned by the proposed method, a t-SNE visualization experiment was conducted on the PU dataset. In particular, the high-dimensional features extracted by the final layer of the network were projected into a 2-D space using t-SNE, where different colors denote different classes. As shown in Fig. 9, the features generated by the proposed method exhibit superior clustering characteristics: samples from the same class are distributed more compactly, indicating significantly enhanced intraclass compactness, while the boundaries between different classes become clearer, leading to improved interclass separability. In particular, for land-cover categories with highly similar spectral characteristics that are prone to confusion, the proposed method still demonstrates strong discrimination ability in the low-dimensional embedding space. These results suggest that the proposed method is capable

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of learning more discriminative deep feature representations, thereby confirming its effectiveness and robustness for HSI classification on the PU dataset.

J. Quantitative Results and Analysis

Tables III–V present the classification results of all compared methods on the PU, SAs, and HongHu datasets, including OA, AA, the Kappa coefficient, and per-class classification accuracies, where the best results are highlighted in bold. It can be observed from Tables III to V that, compared with the competing methods, the proposed HSMMamba achieves superior classification performance on all three datasets, with particularly prominent advantages in terms of OA, AA, and Kappa, demonstrating its strong overall effectiveness.

K. Visualization Results and Analysis

To provide a more intuitive evaluation of the classification performance of the proposed HSMMamba on different hyperspectral remote sensing datasets, visualization-based analysis was conducted for the classification maps produced by all compared methods. Figs. 10–12 present the classification results of different methods on the PU, SAs, and HongHu datasets, respectively. It can be observed from these figures that, compared with the competing methods, the classification maps generated by HSMMamba exhibit clearer boundaries, stronger class consistency, and higher overall classification accuracy. Moreover, the obtained results are more consistent with the ground-truth land-cover distribution, which further verifies the effectiveness and superiority of the proposed method for HSI classification.

IV. Conclusion

To address the loss of local fine-grained information, the difficulty of adaptive multiscale modeling, and the inadequate modeling of long-range spatial–spectral dependencies in HSI classification, this article proposes HSMMamba, a unified framework that integrates hierarchical superpixels with Mamba. In particular, a LHSM mechanism is introduced to adaptively optimize cross-level mappings while preserving structural priors, a hierarchical scale-aware mixture-of-experts mechanism is developed to enhance heterogeneous features at different scales, and an FBDMamba module is designed to jointly capture global dependencies and local variations. Through the collaboration of these components, HSMMamba effectively exploits multiscale spatial structures and spectral discriminative information within a unified framework.

Extensive experiments demonstrate that HSMMamba consistently outperforms existing mainstream methods in terms of OA, AA, the Kappa coefficient, and visualization quality, while showing stronger robustness in complex scenes and boundary regions. Ablation studies further confirm that the three key components are highly complementary and jointly contribute to performance improvement. Overall, HSMMamba provides an effective solution for HSI classification by jointly considering global dependencies, local details, and multiscale structural modeling. Future work will further investigate its generalization ability on larger scale datasets, as well as its potential for real-time inference and lightweight deployment.

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